



Multimodal route choice models of public transport passengers in the Greater Copenhagen Area

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Received: 31 January 2014 / Accepted: 29 September 2014 / Published online: 15 October 2014
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Abstract Understanding route choice behavior is crucial to explain travelers' preferences and to predict traffic flows under different scenarios. A growing body of literature has concentrated on public transport users without, however, concentrating on multimodal public transport networks because of their inherent complexity and challenges. In particular, choice set generation and modeling route choice behavior while accounting for similarity across alternatives and heterogeneity across travelers are non-trivial challenges. This paper tackles these challenges by focusing on the revealed preferences of 5,641 public transport users in the Greater Copenhagen Area. A two-stage approach consisting of choice set generation and route choice model estimation allowed uncovering the preferences of the users of this multimodal large-scale public transport network. The results illustrate the rates of substitution not only of the in-vehicle times for different public transport modes, but also of the other time components (e.g., access, walking, waiting, transfer) composing the door-to-door experience of using a multimodal public transport network, differentiating by trip length and purpose, and accounting for heterogeneity across travelers.

Keywords Route choice · Public transport · Multimodal networks · Choice set generation · Model estimation · Mixed path size correction logit

1 Introduction

Understanding route choice behavior is crucial to explain preferences of travelers and to forecast traffic flows given future scenarios. The literature in route choice

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modeling has largely focused on the behavior of car drivers (see, for a review, Prato 2009), rather than the one of public transport users. Notably, identifying the relevant factors affecting route choice decisions in public transport networks could guide stakeholders (e.g., local government agencies, public transport agencies) toward the improvement of public transport networks that are more responsive to the user preference structures.

This study tackles the challenge of modeling the behavior of public transport users in a multimodal large-scale network by estimating route choice models from 5,641 observed routes collected in the Greater Copenhagen Area. The multimodal nature of the network, its grid composition and its extension make the modeling effort much more challenging when compared to hub-and-spoke networks where buses only serve as feeder modes to trains.

A traditional two-stage approach to route choice modeling was taken in this study: (1) a doubly stochastic approach for choice set generation was adapted to the public transport context to create plausible alternatives to the observed choices; (2) a mixed path size correction logit was proposed to account for similarity across routes and heterogeneity across travelers. The utility function was specified to consider the entire door-to-door experience in the multimodal network in terms of travel time components: access, egress, waiting, in-vehicle and transfer. Moreover, trip length, trip purpose and travelers' characteristics provided a richer insight into the preference structures of different travelers with different motivations for traveling.

This paper is the first to consider revealed preferences (RP) of public transport users in a large-scale multimodal network where choice set generation and route choice model estimation challenges are solved for the inherent complexity of the problem. Moreover, this paper provides insight into the rates of substitution of the entire door-to-door experience in the multimodal network and compares findings with previous studies focusing either on stated preferences (SP) data or RP data from regional or hub-and-spoke networks.

The remainder of the paper is organized as follows. Section 2 presents the context of the route choice literature on public transport. Section 3 illustrates the data and Sect. 4 introduces the methods used in this study. Section 5 shows the model estimation results and Sect. 6 discusses and elaborates upon these results. Section 7 presents the conclusions of this study.

2 Literature on public transport route choice

For the last 20 years, a limited body of literature has explored the preferences of public transport users. Several contributions to previous literature on public transport route choice used data collected from SP surveys (Nielsen et al. 2001; Vrtic and Axhausen 2002; Fosgerau et al. 2007a, b; Nielsen and Johansen 2012). The drawback of using SP data is that respondents are not actually experiencing the actual trip and the sensitivity to the determinants of route choice preferences may be biased accordingly. The advantage of using RP surveys consists in obtaining greater insight into actual choices of travelers, but RP studies are expensive to carry out and

far less used in literature. The current study overcomes this limitation by proposing an efficient RP data collection within traditional travel survey methods, hence allowing to model actual route choices in a large and complex multimodal public transport network.

A few studies worked with hub-and-spoke networks and therefore only considered very schematic and limited networks. Van der Waard (1988), Vrtic and Axhausen (2002) and Bovy and Hoogendoorn-Lanser (2005) focused on train corridors, only considering buses as access and egress modes, while Raveau et al. (2011) examined the metro network in Santiago (Chile) and considered the routes between metro stations only. Considering hub-and-spoke networks limits the variety of the alternative routes and consequently reduces the possibility of estimating real rates of substitution between different public transport modes. Eluru et al. (2012) applied their model in a complex network but generated a limited number of alternative routes which reduced the variety in the choice set. The current study challenges the previous literature by generating choice sets in a large and complex multimodal public transport network via the adaptation of the doubly stochastic approach to the public transport network.

Several route choice models presented in the literature were multinomial logit models (MNL) (Van der Waard 1988; Bovy and Hoogendoorn-Lanser 2005; Raveau et al. 2011; Fosgerau et al. 2007a, b) which do not take similarity between alternatives into account. Bovy and Hoogendoorn-Lanser (2005) used a nested logit (NL) model only because of the hub-and-spoke structure of their examined network. Bovy and Hoogendoorn-Lanser (2005) were an exception and took similarity among alternatives into consideration by estimating path size logit and C-logit models. Finally, Eluru et al. (2012) estimated mixed logit models without considering path similarities, most likely because the mixed term actually captures not only taste heterogeneity, but also alternative similarity. We expanded the literature on public route choice models by proposing a mixed path size correction logit model considering both the similarities among routes and the heterogeneity among travelers, and disentangling the former in the deterministic part of the utility function and the latter in the random distribution of the preference parameters.

The rates of substitution between travel time components for the presented studies are summarized in Table 1, alongside with the evidence by Tørset (2005) and Abrantes and Wardman (2011) collected from travel surveys and from SP and RP studies to estimate route choice attributes and overall time multipliers compared to in-vehicle time.

Overall, it should be noted that the presented literature has limitations in the data (SP studies) and, when RP data are considered, limitations apply to the extension of the network, the modes considered or the alternative routes generated in complex networks.

Table 2 presents the data and network characteristics of the public transport route choice models in terms of data, networks and model specifications applied, where this paper adds to the literature by presenting a public transport route choice model estimated on RP data, in a large and complex network using model formulations with overlap and mixing models including overlap.

Table 1 Rates of substitution travel time components in the literature on public transport route choice

Parameter/traveler group	van der Waard (1988) All	Nielsen et al. (2001)		Vrtic and Axhausen (2002)			Bovy and Hoogendoorn-Lanser (2005) All	Torset (2005) Commuter
		Commuter	Business	Education leisure	Commuter business	Shopping	Leisure	
Access time	2.3	1.3	0.9	1.7			0.8–1.6	0.4
Egress time	1.2	1.3	0.9	1.7				0.4
Hidden waiting time		0.5	0.3	0.6				
Waiting time at first stop	1.4						2.2	
In-vehicle time	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
(scaled to mode)	(all modes)	(bus)	(bus)	(bus)	(train)	(train)	(train)	(all modes)
In-vehicle time Bus	0.9–1.0							
In-vehicle time Metro		0.8	0.8	0.8				
In-vehicle time S-train			0.7	0.8				
In-vehicle time Train		1.9						
In-vehicle time IC (>60 min)		1.1	0.9	2.3			1.9	
Walking time at transfer	2.2	1.1	0.9	2.3			2.2	4.3
Waiting time at transfer	1.2	1.1	0.9	2.3				
Transfer time					0.6–0.8	1.2		
Number of transfers	5.9	15.1	13.5	20.0	7.6–8.6	6.0	5.1–11.4	
Headway					0.0–0.3	0.2		
Delay at the destination		1.4	1.4	2.7				
Penalty if no seat		4.5	4.0	15.0				
Other								

Table 1 continued

Parameter/traveler group	Fosgerau et al. (2007a)			Fosgerau et al. (2007b)		Abrantes and Wardman (2011) All	Raveau et al. (2011) All	Eluru et al. (2012) All (st.dev.)	Nielsen and Johansen (2012)		
	Bus	Metro	S-train	Train	Metro	Train			Commuter	Business	Leisure
Access time	1.8	1.3	1.3	1.3	1.1	1.0			1.1	1.0	1.1
Egress time	1.8	1.3	1.3	1.3	1.1	1.0			1.1	1.0	1.1
Hidden waiting time	0.5	1.3	0–9	0.4	0.6	0.5			0.5	0.5	0.6
Waiting time at first stop											
In-vehicle time	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
(scaled to mode)	(bus)	(metro)	(S-train)	(train)	(bus)	(bus)	(all modes)	(bus)	(bus)	(bus)	(bus)
In-vehicle time Bus											
In-vehicle time Metro					0.8	0.8		0.6	0.7	0.7	0.7
In-vehicle time S-train					0.8	0.8			0.7	0.7	0.7
In-vehicle time Train					0.8	0.8		0.7 (0.2)	0.9	0.9	0.9
In-vehicle time IC (>60 min)											
Walking time at transfer	2.0	2.5	1.9	1.4	1.1	1.4	1.7	1.3	1.1	1.0	1.1
Waiting time at transfer	2.0	2.5	1.9	1.4	1.1	1.4	1.7	0.28/tr.	1.1	1.0	1.1
Transfer time	12.2	1.6	5.2	12.9	8.5	5.5					
Number of transfers								9.3 (3.6)	4.0	4.0	6.0
Headway							0.8				
Delay at the destination											
Penalty if no seat											
Other						1.4		–2.4 to 5.8			

Table 2 Characteristics of the data, the networks and the model specifications in the literature of public transport route choice models

Data/network	Model specification			
	MNL	NL	Overlap	Mixing Mixing and overlap
SP/hub-and-spoke	Vrtic and Axhausen (2002)			
RP/hub-and-spoke	Van der Waard (1988)	Bovy and Hoogendoorn-Lanser (2005)	Bovy and Hoogendoorn-Lanser (2005)	Eluru et al. (2012)
	Bovy and Hoogendoorn-Lanser (2005)			
	Raveau et al. (2011)			Nielsen et al. (2001)
SP/network	Nielsen et al. (2001)			Fosgerau et al. (2007a, b)
	Fosgerau et al. (2007a, b)			Nielsen and Johansen (2012)
RP/network			Anderson et al. (present study)	Anderson et al. (present study)

This study tackles the challenges of modeling route choice in multimodal public transport networks with a two-stage approach that implies choice set generation and discrete choice model estimation on an inherently complex transport system.

3 Data

The multimodal public transport network of the Greater Copenhagen Area was used for the analysis in the study. The public transport services in the area include metro, buses, and trains (regional, IC-trains, S-trains and local trains). Buses near the central business district (CBD) of Copenhagen have 3–10 min frequencies, while buses in the outskirts and in the rural areas have 20–60 min frequencies: metro 2 min, S-train 5–10 min and regional and IC-trains' frequencies from 20 to 120 min (some even less often). The area is the most densely populated area in Denmark with around 2 million residents and the most important characteristics of the public transport network are:

- The CBD of Copenhagen is served by the metro and A-buses.
- The S-trains serve the “fingers” to the other great cities in the area.
- The fast buses connect the S-train lines by driving in rings around the city of Copenhagen.
- The local trains mainly serve the northern part of the region.
- The regional and IC-trains serve the legs leading from the area to other parts of Denmark.
- The remaining buses serve all the remaining areas, often also serving a train station.

Data were collected via the TU survey and information about the Greater Copenhagen Area was represented via a GIS network that was used for route map-matching.

The TU Survey is an on-going collection of travel diaries alongside respondents' and households' socioeconomic data and consists of a questionnaire which is either answered on the internet (20 %) or via telephone (80 %). In the TU survey, respondents are a representative sample of the Danish population between 10 and 84 years who are asked to describe all their trips with both private and public transport modes on the day before the interview (for details see Christensen 2013). Respondents provide information on all their trips during the day (e.g., selected modes, time duration, distance traveled, and trip purpose) and all their sociodemographic characteristics (e.g., gender, age, income, location of residence and workspace).

One peculiarity of the TU survey is the method used for collecting detailed information about route choices of public transport travelers (for details see Anderson 2010, 2013). The collected information is detailed enough to enable the matching of the route to a GIS network, but simple enough for the respondent to provide it seamlessly and correctly. In general, for each trip, the respondent is asked

for starting time, origin location, end time, destination location, purpose of the stay and transport modes used for the trip. All transport modes used on the trip are asked for. The respondent chooses from a drop down list (21 transport modes including car, bicycle, walking, bus, train). When choosing a mode, additional boxes are to be filled according to the transport mode entered: length and travel time (walking, bicycle, car, airplane); waiting time, bus line, length and travel time (bus); waiting time, from-station, S-train line, to-station, length and travel time (suburban S-train); waiting time, from-station, to-station, length and travel time (metro and train). Figure 1 shows an example of the description of a public transport route.

The collected data were map-matched to the GIS network by identifying the actual public transport lines, bus stops, train stations and schedules used by the traveler (for details see Anderson and Rasmussen 2010). The GIS network is a schedule-based network and consists of 369 public transport lines with a total of 18,487 runs, 292 stations and 4,729 bus stops, with 1,718 between-mode transfer options (multimodal transfers). The respondent's departure time and route were map-matched to the public transport schedules and the times and distances from the model were used for the remaining study for consistency.

From February 2009 to May 2010, approximately 6,300 interviews were collected in the Greater Copenhagen Area with more than 22,500 trips of which 2,200 used public transport for a part of the trip (minimum one trip leg). The map-matching algorithms provided a successful map-matching of 91 % of the observed trips. The remaining 9 % of the trips mainly consisted of observations with missing or incorrect information on bus line use. This information had to be precise and most often it could not be corrected manually and the observation had to be discarded. The map-matched dataset of 5,641 trips was processed and used in this paper.

Transport modes in correct order:					
	Transport mode	Line	Length	Time	
1.	Walking		0.5 km	8 min	
			Waiting time:	3 min	
2.	Bus	701	2 km	5 min	
3.	Walking		0.2 km	2 min	
	From-station: Nykøbing F		Waiting time:	7 min	
4.	Other train		146 km	100 min	We calculated the distance to 146 km from the stated stations
	Transfer at: København H		Waiting time:	8 min	
5.	S-train	E	13.9 km	20 min	We calculated the distance to 13.9 km from the stated stations
	To-station: Lyngby		Waiting time:	5 min	
6.	Bus	300s	5 km	10 min	
7.	Walking		0.5 km	5 min	
	Sum:		168.1 km	173 min	

Fig. 1 Route choice questions in the travel survey—trip description

4 Model

4.1 Choice set generation

Based on the map-matched routes, attractive routes were generated for the origin–destination pair of each traveler for model estimation purposes. A doubly stochastic path generation algorithm (Nielsen 2000) was adapted to the public transport context to generate routes relevant to the travelers. The method allows for both variation in the perceived costs of the network elements and heterogeneity in the preferences of the travelers. The coverage of the observed choices with the generated choice sets provides a measure of the performance of the applied route choice generation technique. The coverage is measured in percentage overlap with the observed route and in this particular context is calculated at two levels: stop level and link level. The link level is the most restrictive measure and the examination of the coverage will be considered for the generated choice sets when comparing public transport lines on the same links.

The doubly stochastic path generation algorithm is based on a schedule-based stochastic transit assignment model solved with an MSA and proposed by Nielsen (2000), further improved by Nielsen (2004), and finally refined by Nielsen and Frederiksen (2006). Error Components are included within the utility function to express different preferences across different individuals:

$$U_l = \sum_n (\beta_n + \xi_n) t_{ln} + \varepsilon_l, \quad (1)$$

where U_l is the utility of link l , t_{ln} are time components on link l as perceived by individual n , β_n is the coefficient expressing individual preferences for a certain time component, ξ_n is the error component expressing variation in these individual preferences, and ε_l is a Gamma distributed error term for link l . The Gamma distribution allows avoiding negative link costs and summing over the links, since the distribution is additive. This property is desirable when aggregating from the link to the route level, as the utility of any alternative route is given by the sum of the utility of its links, and hence Eq. (1) is valid at both the link and the route levels. For the Gamma distribution, we assume a constant relation between mean and variance of the utility, as explained by Nielsen and Jovicic (1999), whereby the distribution is described by only one parameter.

Larsen et al. (2010) considered three different formulations for the utility function, to test the effect of the variation of the different time components, and six values of the variances of the distributions of both error components and error term. The three formulations were defined as:

- ErrTermOnly: all the β 's are fixed and the ε_l is randomly distributed according to a Gamma distribution.
- ErrCompAll: all the β 's are randomly distributed according to a log-normal distribution, and the ε_l is not considered in the utility function.

- ErrCompErrTerm: all the β 's are randomly distributed according to a log-normal distribution, and the ε_l is randomly distributed according to a Gamma distribution.

In expression (1), the mean VoT (β_n) and the variance of the log-normal distribution are given, leading to obtaining the parameters of the distribution.

For each of the three utility function formulations, the error term was considered with six different values of the scale parameter (0.025, 0.05, 0.1, 0.2, 0.5, 1.0), and the VOT terms were examined with six different variance scale parameters (0.05, 0.10, 0.15, 0.2, 0.5, 1.0).

Routes were generated by simulating the utility functions for 200 times for each of the six scenarios (i.e., one for each scale parameter value) of the three formulations. The routes were produced as sequences of links within the network and then compared to the observed routes to calculate overlap and coverage.

4.2 Route choice model estimation

Given the generated choice sets C_n for each observation n , we proposed a route choice model by assuming that travelers maximize their utility of choosing alternative k from C_n :

$$U_{kn} = V_{kn} + \varepsilon_{kn} \quad \forall k \in C_n, \quad (2)$$

where V_{kn} is the deterministic component and ε_{kn} is the random component of the utility.

The random component of the utility was assumed to be Gumbel distributed, while the deterministic component of the utility was specified as follows:

$$V_{kn} = \sum_m \beta_{ivt,m} ivt_{mkn} + \sum_c \beta_{t,c} t_{ckn} + \sum_q \beta_{pen,q} pen_{qkn} + \beta_{PSC} PSC_{kn}, \quad (3)$$

where ivt_{mkn} are in-vehicle travel times for public transport mode m used in route k by traveler n , t_{ckn} are other time components c (e.g., walking, waiting, transfer time, transfer waiting, transfer walking) of the door-to-door multimodal public transport route k of traveler n , pen_{qkn} is the penalty of one transfer type q (e.g., transfer between all modes or between specific modes bus, trains), PSC_{kn} is the path size correction term for traveler n (Bovy et al. 2009), and the corresponding β 's are parameters to be estimated. It should be noted that fare cannot be entered in the model because of the zonal fare system in the Greater Copenhagen Area that makes every trip connecting the same origin and the same destination cost the same independent on the combination of public transport modes used.

Accordingly, the deterministic part of the utility is defined as the sum of time components and the path size correction term that accounts for the similarity across the alternatives. Namely, this correction term reduces the utility of alternative routes that are similar to other routes in the choice set according to the following expression:

$$\text{PSC}_{kn} = - \sum_{a \in \Gamma_k} \left(\frac{L_a}{L_k} \ln \sum_{h \in C_n} \delta_{ah} \right), \quad (4)$$

where L_a is the length of link a , L_k is the length of route k , Γ_k is the set of links belonging to route k , δ_{ah} is the link-route incidence dummy equal to one if route h uses link a and zero otherwise.

Given the distribution of the error term and the specification of the deterministic utility, the probability P_{kn} of traveler n choosing alternative route k within the choice set C_n is expressed as a PSC-Logit (Bovy et al. 2009):

$$P_{kn} = \frac{\exp(V_{kn})}{\sum_{h \in C_n} \exp(V_{hn})} = \frac{\exp \left(\sum_m \beta_{\text{ivt},m} \text{ivt}_{mkn} + \sum_c \beta_{t,c} t_{ckn} + \sum_q \beta_{\text{pen},q} \text{pen}_{qkn} + \beta_{\text{PSC}} \text{PSC}_{kn} \right)}{\sum_{h \in C_n} \exp \left(\sum_m \beta_{\text{ivt},m} \text{ivt}_{mhn} + \sum_c \beta_{t,c} t_{chn} + \sum_q \beta_{\text{pen},q} \text{pen}_{qhn} + \beta_{\text{PSC}} \text{PSC}_{hn} \right)}, \quad (5)$$

where the closed-form logit-type formula of the PSC-Logit allows for simple estimation of the fixed β 's via maximum likelihood. It should be noted that the PSC-Logit was derived theoretically, as was the original formulation of the PS-Logit (Ben-Akiva and Bierlaire 1999), unlike most of the existing corrections that are heuristic per se.

Alongside the similarity across alternative routes, we intended to consider within the route choice model the heterogeneity across travelers by allowing the β 's to vary across the observations and hence estimating a Mixed PSC-Logit model. Given the density distribution of the parameters in the utility function, the probability P_{kn} of traveler n choosing alternative route k within the choice set C_n is expressed as:

$$P_{kn} = \int \frac{\exp(V_{kn})}{\sum_{h \in C_n} \exp(V_{hn})} f(\beta_{\text{ivt},m}) f(\beta_{t,c}) f(\beta_{\text{pen},q}) d\beta_{\text{ivt},m} d\beta_{t,c} d\beta_{\text{pen},q}, \quad (6)$$

where the probability is integrated over the distribution of the β 's. As the integral is multi-dimensional, the following expression is maximized via simulation:

$$\text{SLL} = \sum_{n=1}^N \sum_{h \in C_n} d_{hn} \ln \left\{ \frac{1}{R} \sum_{r=1}^R \left[\frac{\exp \left(\sum_m \beta_{\text{ivt},m}^r \text{ivt}_{mkn} + \sum_c \beta_{t,c}^r t_{ckn} + \sum_q \beta_{\text{pen},q}^r \text{pen}_{qkn} + \beta_{\text{PSC}} \text{PSC}_{kn} \right)}{\sum_{h \in C_n} \exp \left(\sum_m \beta_{\text{ivt},m}^r \text{ivt}_{mhn} + \sum_c \beta_{t,c}^r t_{chn} + \sum_q \beta_{\text{pen},q}^r \text{pen}_{qhn} + \beta_{\text{PSC}} \text{PSC}_{hn} \right)} \right] \right\}, \quad (7)$$

where SLL is the simulated log-likelihood, N is the number of travelers, d_{hn} is equal to 1 if traveler n has selected alternative route h and 0 otherwise, r is one of the R random draws required for integral simulation, and the superscript r represents the instance of a draw for the parameters to be estimated. According to Nielsen (2000),

log-normal distributions are preferred for the in-vehicle travel time and the other time components. This formulation prevents confounding the similarity across the alternative paths (represented within the deterministic part of the equation) from the taste heterogeneity (accounted for in the distributed parameters).

In the present study, the parameters were estimated using 500 random draws from a Modified Latin Hypercube Sampling (MLHS) method (Hess et al. 2006). The selection of the MLHS method over the more commonly used Halton draws is justified by the fact that correlation patterns would arise across Halton draws for the multi-dimensional integral with consequent possible bias in the parameter estimates, and by the fact that MLHS random draws outperform alternative sampling methods in multi-dimensional integrals (Hess et al. 2006). The freeware software BIOGEME (Bierlaire 2009) was used for model estimation thanks to its simplicity and versatility in specifying the model formulated for this analysis.

5 Results

Using the generated alternatives, the final choice sets for estimation purposes were created performing the following steps:

1. Generate route choice set R_{all} containing 200 routes from O to D.
2. Identify unique routes in choice set R_{all} and create new choice set R_{unique} containing only unique routes. A unique route is unique at line (or stop) level which means that at least one link (or stop) has to be different from the other routes in the choice set.
3. Validate the route choice set R_{all} with actual chosen routes (coverage = 100 % if the complete route k is in the choice set R_{all}).
4. If the observed route k is not already in the choice set R_{unique} add k to R_{unique} to create final choice set R_{final} .
5. Use final choice set R_{final} to estimate route choice models.

5.1 Choice set generation

For each generated route h , the overlap $O_{h,lv,n}$ of each generated path h with the observed route of observation n was measured as follows (Ramming 2002):

$$O_{h,lv,n} = \frac{L_{hn}}{L_n}, \quad (8)$$

where L_{hn} is the sum of the length of overlapping elements at level lv between the generated and the observed route for traveler n and L_n is the sum of the attributes of the observed route. The elements are evaluated at the line and at the stop level. At the line level, these elements correspond to the public transport lines, which means that a match is found only when the same line is used for both the observed and the generated route. At the stop level, these attributes correspond to the stop

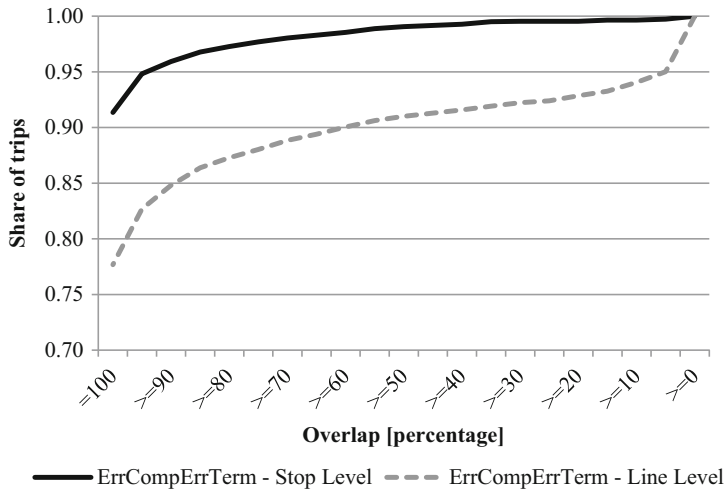


Fig. 2 Overlap and coverage at stop and line level for the ErrCompErrTerm utility formulation with scale parameter 1.0

connections, which means that lines using the same links and stops are considered to be matching elements.

This overlap measure can be computed for each generated path h for observation n , and let $O_{\max\{n\},lv,n}$ denote the best overlap (measured at level lv using overlap-in-length as unit of measure) among the paths generated for observation n . Then the coverage for an overlap threshold equal to δ can be computed as (Ramming 2002):

$$\text{Cov}_{lv}(\delta) = \frac{\sum_{n=1}^N I(O_{\max\{n\},lv,n} \geq \delta)}{N}, \quad (9)$$

where N is the number of observations and $I(\cdot)$ is an indicator equal to 1 when the criterion is fulfilled and 0 otherwise.

Figure 2 shows the coverage for the scenario with the highest scale parameter (1.0). The best coverage produced by the doubly stochastic path generation method (the ErrTermErrComp) with these parameters indicates high consistency between the choice set generation and the observed behavior. Accordingly, this parameterization is preferable to avoid sample bias and to strive toward consistent estimation results. By visually inspecting the generated choice sets we found that with increasing scale parameter also the counterintuitive routes (e.g., route with loops) increased and therefore the level of 1.0 for the scale parameters was retained.

The choice set generated by this parameterization is used for model estimation and the total number of unique routes over all observations n was equal to 226,845 alternatives, with a minimum of 2, a maximum of 96, and an average of 40.4 alternatives per choice set. The number of unique routes depends on the possible routes of the OD pair. Often an OD pair for a short trip in a rural area will have few unique routes in the generated choice set and an OD pair in a high level-of-service

public transport area will have a high number of unique routes in the choice set. The exact same route can be simulated numerous times but is only kept once in the final choice set, R_{final} .

5.2 Route choice models

Considering the observations over the 80 % overlap threshold as consistent with the observed behavior, MNL, PSC-Logit and Mixed PSC-Logit models were estimated for the overall sample containing 5,641 observations. Estimates allowed calculating rates of substitution between in-vehicle times for different modes m and time components of the door-to-door trip on the multimodal public transport network.

A large number of model specifications were formulated testing different parameters for in-vehicle time, other time components, mode specific parameters, transfer characteristics, etc. The process was made iteratively, the result of one estimation leading to the formulation of the following model. For certain parameters following a piece-wise function the threshold values were decided with the same trial-and-error approach. All model specifications and combinations of parameters were estimated both as MNL and PSC-logit models. The PSC-logit models systematically outperformed the MNL models by giving better model fit and higher log-likelihood values. Below we present a selection of the best performing and most interesting models.

5.3 PSC-Logit by trip purpose

Trip purposes were reported in the TU survey, and specification testing allowed differentiating between work-related trips (2,952 observations) and leisure-other trips (2,689 observations). The work-related trips included commuting trips, trips to educational institutions and business trips, while the leisure-other trips covered the remaining purposes.

The initial specification of the utility function for the estimation of the PSC-Logit considered in-vehicle travel time for bus, metro, S-trains, regional/IC-trains and local trains. Also, time components included access/egress time, walking time, waiting time at stops and transfer penalties.

Table 3 shows the estimated parameters and the rates of substitution when considering coefficient values for the estimations by purpose compared to the estimation with all trips.

The estimated parameters are significant and with the expected sign, with the exception of the metro parameter and the path size parameter. The metro parameter is not significantly different from zero for leisure trips. The path size correction is intuitively expected to be positive since it, in principle, implies a reduction of the utility for highly overlapping routes, with this reduction being obtained with a positive parameter since the logarithm of a number between 0 and infinite (the extremes of the path size correction) is negative. The log-likelihood for the estimated model is high implying a good model fit. The rates of substitution suggest that leisure travelers perceive transferring as more burdensome than travelers commuting to their workplace: while the latter would rather spend 12 additional min

Table 3 PSC-Logit estimates, (*t* tests) and rates of substitution (w.r.t. bus in-vehicle time) for trip purposes

Parameter	Model estimates		Rates of substitution			
	Trip purpose		Trip purpose			
	All	Work	All	Work	Leisure	Leisure
In-vehicle travel times						
Bus	-0.143 (-31.4)	-0.166 (-25.3)	1.0	1.0	1.0	1.0
Metro	-0.027 (-3.21)	-0.046 (-4.0)	0.2	0.3	–	–
S-train	-0.076 (-14.5)	-0.101 (-13.7)	0.5	0.6	0.4	0.4
Local train	-0.101 (-9.1)	-0.136 (-10.6)	0.7	0.8	0.5	0.5
Regional + IC-train	-0.113 (-13.0)	-0.135 (-11.5)	0.8	0.8	0.7	0.7
Other time components						
Access/egress time	-0.343 (-50.9)	-0.365 (-36.6)	2.4	2.2	2.8	2.8
Waiting time at transfers	-0.083 (-13.7)	-0.082 (-14.1)	0.6	0.5	0.7	0.7
Walking time at transfers	-0.132 (-10.1)	-0.137 (-7.4)	0.9	0.8	1.1	1.1
Number of transfers	-2.020 (-42.0)	-2.020 (-29.7)	14.1	12.2	17.9	17.9
Path size correction						
PSC	-0.755 (-14.0)	-0.754 (-9.8)				
Number of observations	5,641	2,952				
Null log-likelihood	-20,172	-10,724				
Final log-likelihood	-11,760	-6,082				
Adjusted rho-square	0.417	0.432				

on the bus rather than transferring, this value reaches 18 min for the former. Moreover, leisure travelers perceive in-vehicle time on local and S-trains as well as access/egress times as less burdensome than travelers on work-related trips.

5.4 PSC-Logit models for trip purpose and trip length

Trip length was analyzed by dividing the observations into distance bands as suggested in Fosgerau et al. (2007b). Specification tests for various values of these bands revealed that the best model estimates were obtained when considering 10 km as the threshold between short and long trips. Models were then estimated by considering the combination of trip purpose (i.e., work-related vs. leisure-other) and trip length (i.e., less than 10 km, more than 10 km). The same specification of the utility function was maintained, and Table 4 presents the estimated parameters and the rates of substitution.

The estimated parameters have the expected sign and are significantly different from zero, with the exception of the metro parameter not being significant for longer leisure trips and the path size correction not being positive. The rates of substitution confirm the tendencies uncovered when analyzing only the trip purpose and provide insight into comparable rates for in-vehicle travel times between shorter and longer trips. However, differences emerge in the number of transfers, where values for leisure trips and for longer trips are higher. Also waiting and walking times at transfers are perceived as more burdensome (relative to in-vehicle time on buses) for leisure and longer trips, with the exception of longer leisure trips where the waiting time at transfers is found less burdensome than for work-related trips and shorter trips in general.

5.5 PSC-Logit model for trip purpose and headway considerations

Given the comparable rates of substitution for the in-vehicle times when considering different trip lengths, model estimation continued accounting for differences according to trip purposes and moved toward adding time components to the specification of the utility function. Specifically, the headway was considered by taking into account half of the highest headway in the multimodal trip and separating headway up to and exceeding 6 min with a piece-wise linear specification. This value was based on the Danish “rule-of-thumb” that travelers will start by timing their arrival according to the timetable at low-frequency services and arrive randomly at high-frequency services. Some lines (e.g., metro and A-buses) run with such a high frequency (down to 2 min headway) that there is no published timetable.

Similarly, the in-vehicle travel time on regional and IC-trains was considered with a linear piece-wise specification by dividing travel time for length up to and exceeding 20 km. The 20 km threshold is in line with the fact that the average daily commuter distance in the Greater Copenhagen Area is approximately 20 km. Finally, the parameter for number of transfers is exchanged by four transfer parameters which also take the transport modes transferred to and from into consideration.

Table 4 PSC-Logit estimates (*t* tests) and rates of substitution (w.r.t. bus in-vehicle time) for trip purposes and length

Parameter	Model estimates		Rates of substitution			
	Trip purpose		Trip purpose			
	Work		Work		Leisure	
	<10 km	>10 km	<10 km	>10 km	<10 km	>10 km
In-vehicle travel times						
Bus	-0.209 (-17.8)	-0.139 (-18.7)	-0.154 (-16.6)	-0.082 (-11.5)	1.0	1.0
Metro	-0.061 (-3.4)	-0.046 (-3.1)	-0.032 (-1.9)	0.012 (0.7)	0.3	-
S-train	-0.111 (-6.2)	-0.089 (-11.3)	-0.072 (-5.6)	-0.021 (-2.8)	0.5	0.6
Local train	-0.176 (-2.9)	-0.111 (-9.0)	-0.134 (-2.7)	-0.041 (-2.8)	0.8	0.8
Regional + IC-train	-0.223 (-5.0)	-0.113 (-9.5)	-0.174 (-4.6)	-0.053 (-4.5)	1.1	0.8
Other time components						
Access/egress time	-0.427 (-23.8)	-0.320 (-28.0)	-0.356 (-28.1)	-0.269 (-22.3)	2.0	2.3
Waiting time at transfers	-0.128 (-8.4)	-0.063 (-11.0)	-0.219 (-10.8)	-0.035 (-3.6)	0.6	0.5
Walking time at transfers	-0.166 (-5.8)	-0.123 (-5.0)	-0.179 (-6.6)	-0.100 (-4.1)	0.8	0.9
Number of transfers	-2.120 (-19.3)	-1.880 (-22.0)	-1.920 (-22.2)	-1.960 (-18.9)	10.1	13.5
Path size correction						
PSC	-0.773 (-8.0)	-0.715 (-5.8)	-0.825 (-9.2)	-0.583 (-4.17)		
Number of observations:	1,392	1,560	1,793	896		
Null log-likelihood:	-4,951	-5,777	-6,153	-3,293		
Final log-likelihood:	-2,766	-3,265	-3,551	-1,955		
Adjusted rho-square:	0.439	0.433	0.421	0.403		

In Table 5 presents the estimation results for the model the estimated parameters have the expected sign and are significant, with only the path size correction parameter not being positive as anticipated. The rates of substitution show that the headway plays a different role depending on whether it is for more or less frequent services: for short headways in fact, the values are much lower than for long headways. Also, the in-vehicle travel time of regional and IC-trains is perceived as much more burdensome for shorter trips than for longer trips, where the higher speed of these trains is clearly an advantage. Interestingly, for longer trips, these trains become as competitive as the metro. For multimodal transfers, a transfer to a train (including metro) is found less burdensome than a transfer to bus.

5.6 Mixed PSC-Logit model

Given the investigation of the best specification of the utility function, and considering the goodness-of-fit being higher for the models presented in the previous section, heterogeneity across trip purposes was considered by examining distributions for the estimated parameters. Log-normal distributions were considered for the in-vehicle travel time and the other time components, and rates of substitution were calculated by Monte Carlo simulation with 1 million draws from the estimated distributions.

Following specification and significance tests and evaluation of differences across trip purposes, results are presented for the work-related and the leisure-other travelers in Table 6.

For leisure-other travelers, log-normal parameters were estimated and found significant for bus in-vehicle time, regional and IC-trains up to 20 km in-vehicle time, headway exceeding 6 min, and number of transfers between bus and trains and vice versa. For the travelers with work-related purposes also the parameter for transfers between two buses was found significant. Notably, heterogeneity was found in the bus in-vehicle time parameter that was assumed as reference for the calculation of the rates of substitution and resulted in ranges of the rates of substitution that account for differences across travelers in the perception of the various parts of their trips.

6 Discussion

Insights into route choice behavior in a multimodal large-scale public transport network are in the following obtained from the interpretation of the estimated parameters and the rates of substitution in the presented models.

6.1 In-vehicle travel times

All models presented in the literature study in Sect. 2 show that travelers prefer in-vehicle time in the various train modes over bus in-vehicle time, equivalent to the findings in this study.

Table 5 Best PSC-Logit estimates, (*t* tests) and rates of substitution (w.r.t. bus in-vehicle time) for trip purposes

Parameter	Model estimates		Rates of substitution			
	Trip purpose		Trip purpose			
	All		Work	Leisure	All	Work
In-vehicle travel times						
Bus	-0.132 (-29.6)		-0.156 (-23.6)	-0.105 (-17.6)	1.0	1.0
Metro	-0.075 (-8.29)		-0.100 (-7.86)	-0.048 (-3.72)	0.6	0.6
S-train	-0.095 (-17.1)		-0.123 (-15.1)	-0.061 (-8.24)	0.7	0.8
Local train	-0.080 (-8.14)		-0.124 (-9.10)	-0.051 (-4.09)	0.6	0.8
Regional + IC-train						
Up to 20 km	-0.167 (-10.1)		-0.196 (-8.39)	-0.142 (-5.75)	1.3	1.3
Exceeding 20 km	-0.084 (-7.01)		-0.099 (-6.70)	-0.060 (-2.94)	0.6	0.6
Other time components						
Headway						
Up to 6 min	-0.279 (-7.12)		-0.261 (-4.91)	-0.302 (-5.23)	2.1	1.7
Exceeding 6 min	-0.044 (-7.16)		-0.044 (-5.11)	-0.045 (-5.24)	0.3	0.3
Access/egress time	-0.366 (-51.0)		-0.394 (-36.9)	-0.339 (-34.7)	2.8	2.5
Waiting time at transfers	-0.080 (-13.2)		-0.079 (-13.5)	-0.082 (-6.72)	0.6	0.5
Walking time at transfers	-0.146 (-8.74)		-0.142 (-6.20)	-0.159 (-6.49)	1.1	0.9
Bus → bus transfers	-2.560 (-32.3)		-2.660 (-21.9)	-2.470 (-23.5)	19.4	17.1
Bus → train transfers	-1.890 (-20.1)		-1.980 (-15.4)	-1.810 (-12.8)	14.3	12.7
Train → bus transfers	-2.090 (-22.0)		-2.220 (-17.0)	-1.950 (-13.6)	15.8	14.2
Train → train transfers	-1.180 (-19.7)		-1.120 (-14.0)	-1.290 (-14.0)	8.9	7.2
Path size correction						
PSC	-0.680 (-12.2)		-0.673 (-8.30)	-0.687 (-8.94)		
Number of observations	5,641		2,952	2,689		
Null log-likelihood	-20,172		-10,724	-9,442		
Final log-likelihood	-11,526		-5,935	-5,555		
Adjusted rho-square	0.428		0.445	0.410		

Table 6 Mixed PSC-Logit estimates, (*t* tests), rates of substitution (w.r.t. bus in-vehicle time) and [95 % confidence intervals]

Parameter	Model estimates		Rates of substitution	
	Trip purpose		Trip purpose	
	Work	Leisure	Work	Leisure
In-vehicle travel times				
Bus (μ)	-1.770 (-55.8)	-2.180 (-42.6)	1.0 [1.0–1.0]	1.0 [1.0–1.0]
Bus (σ)	0.193 (5.1)	0.482 (9.4)		
Metro	-0.113 (-9.1)	-0.063 (-5.2)	0.7 [0.5–0.8]	0.6 [0.3–1.0]
S-train	-0.139 (-22.0)	-0.075 (-9.8)	0.8 [0.5–1.1]	0.7 [0.4–1.2]
Local train	-0.160 (-5.2)	-0.085 (-3.36)	1.0 [0.7–1.2]	0.8 [0.4–1.4]
Regional + IC-train				
Up to 20 km (μ)	-1.530 (-20.4)	-1.850 (-13.3)	1.5 [0.8–2.1]	1.9 [0.5–4.0]
Up to 20 km (σ)	0.356 (2.4)	0.663 (3.6)		
Exceeding 20 km	-0.111 (-9.3)	-0.067 (-4.1)	0.7 [0.6–1.0]	0.7 [0.3–1.1]
Other time components				
Headway				
Up to 6 min	-0.224 (-5.1)	-0.283 (-6.2)	1.3 [1.0–1.7]	2.8 [1.3–4.6]
Exceeding 6 min (μ)	-2.950 (-26.2)	-3.080 (-24.0)	0.6 [0.1–1.4]	0.7 [0.1–1.5]
Exceeding 6 min (σ)	1.170 (8.4)	0.917 (4.7)		
Access/egress	-0.438 (-52.8)	-0.372 (-50.9)	2.6 [2.0–3.3]	3.7 [1.8–6.1]
Walking time in transfers	-0.152 (-6.2)	-0.179 (-5.44)	1.2 [0.7–1.6]	1.8 [0.9–2.9]
Waiting time in transfers	-0.083 (-19.7)	-0.086 (-30.5)	0.6 [0.4–0.9]	0.9 [0.4–1.4]
Bus $\rightarrow$ bus transfers (μ)	-1.080 (-8.8)	-2.630 (-24.7)	18.8 [10.1–29.5]	26.2 [12.5–43.2]
Bus $\rightarrow$ bus transfers (σ)	0.369 (12.7)			
Bus $\rightarrow$ train transfers (μ)	-0.714 (4.7)	-0.585 (7.21)	14.2 [5.7–25.3]	21.8 [5.7–44.0]
Bus $\rightarrow$ train transfers (σ)	0.550 (15.5)	0.632 (3.77)		
Train $\rightarrow$ bus transfers (μ)	-0.858 (5.5)	-0.702 (9.29)	17.3 [5.9–32.4]	26.5 [5.7–55.8]
Train $\rightarrow$ bus transfers (σ)	0.636 (14.0)	0.747 (4.12)		
Train $\rightarrow$ train transfers	-1.200 (-19.7)	-1.350 (-13.6)	7.2 [5.5–9.0]	13.4 [6.4–22.2]
Path size correction				
PSC	-0.695 (-12.2)	-0.723 (-12.8)		
Number of observations	2,952	2,689		
Null log-likelihood	-10,724	-9,942		
Final log-likelihood	-5,863	-5,506		
Adjusted rho-square	0.453	0.417		

The metro in Copenhagen has a high frequency (2 min, headway during rush hours) and a high stability of operations (it is automatic and driverless). It was perceived very favorable, even when correcting the models for the frequency and the guarantee of having a fast transfer. A reason for this may be the stability of operations, compared to the other rail products and buses in the CBD of Copenhagen.

The S-train in the Greater Copenhagen Area serves a much larger area than the metro and presents comparable positive perception of the in-vehicle time with respect to the bus. S-train lines constitute the backbone of the transit-oriented development plan from the suburbs to Copenhagen, also known as “the finger plan” from 1947 (see Knowles 2012). The S-trains are comfortable, have free Wi-Fi and the operation is quite punctual (94 % of the trains were less than 2 min delayed in 2010).

Local trains are generally preferred over regional and IC-trains, although they may be considered a bit less comfortable. The advantage of these trains is their high punctuality (97 % less than 5 min delayed in 2012).

Regional and IC-trains are considered less favorably in terms of in-vehicle time, although factoring the length of the trip reveals that for trips over 20 km this is not observed. Only 89.7 % of the trains were less than 6 min delayed in 2010, whereas at the coastal railway (one of the main lines in the survey area) only 81.3 % were less than 5 min delayed in 2010, with large seasonal variations from a monthly average of 60.7–90.5 %. Some of the regional trains use old rolling stock and coaches (compared to the other products), in the public known with the nickname “cattle trucks”, which may also explain their lower preference.

A general conclusion is thus that the differences in perception of time for different trains may be related to comfort and/or punctuality. Unfortunately, information on punctuality is only reported per train operator, but with different threshold values and not for specific lines and day of operations. It was therefore not possible to include this information directly in the disaggregate model estimation.

6.2 Other time components

The explanatory power of the model increased when considering the headway for the public transport mode with the lowest frequency along the route. Factoring the frequency resulted in a small increase in the negative perception related to traveling on regional and IC-trains as well as S-trains, most likely because the transport mode with the highest headway on multimodal trips is often the bus. In the multimodal network of the Greater Copenhagen Area, travelers often use buses in the beginning or the end of the trips to reach the most attractive modes, and this affects the model when factoring this frequency. Significant differences are observed between more or less frequent services, most likely since travelers may time their arrivals in case of low-frequency services whereas they arrive randomly in case of high frequent services which do not have fixed timetables. The few models in the literature estimating headway parameters found values in the range of 0.2–0.8, in line with the estimates in this study for the headways above 6 min.

Access/egress times are generally valued as 2–3 times more burdensome than bus in-vehicle times. The size of the parameter for access/egress time relative to bus is higher than the majority of the literature presented in Sect. 3. It should be noted that, among the literature compared, studies defining buses as access/egress modes are also present.

Transfers are obviously one of the most problematic elements of multimodal trips. Waiting and walking times in transfers are estimated as being burdensome to

travelers, with little variation along the utility specification changes. The penalties for transfers are interestingly high, and the model estimates reveal significant differences in transferring between different modes. Transferring to trains appears less burdensome than transferring to buses, most likely because of their higher reliability and frequency since metro and S-trains are considered alongside the other trains. Also, heterogeneity in the values shows that very different perceptions across travelers exist with respect to the transfer penalty, with values ranging from as little as 5.9 min in the left bound of the confidence interval for bus-to-train transfers to as much as 32.4 min in the right bound of the confidence interval for train-to-bus transfers for work-related travelers. Notably, the transfer penalties are higher for longer trips, possibly because of the possibility to use the in-vehicle time for other activities (e.g., sleeping, reading, working) during longer trips rather than transferring between two shorter trips. Although the number of transfers is obviously higher for longer trips, this tendency illustrates how travelers on longer trips prefer simply to stay on-board and enjoy for example Wi-Fi connection and news screens in trains and fast buses. The fare is not an issue when transferring, since it depends only on the location of the origin and destination in the zone system and the same ticket is used for all public transport modes in the area.

An explanation for the high transfer penalties could be a somewhat lexicographic choice process, i.e., if the choice set has a number of one and two transfer routes, respectively, the traveler only chooses between the one-transfer routes except in the case that the two transfer routes are much faster.

The median values of transfer penalties estimated in this study are in the high end compared to the presented literature, but do not exceed the highest values found in Vrtic and Axhausen (2002) and Nielsen O et al. (2001).

6.3 Path size correction

The parameter for the PSC is significantly different from zero, but it is always negative contrary to the expected positive value that would suggest that the utility of the alternative routes decreases when they overlap with the competitive routes in the choice set. Positive parameters for path sizes are generally obtained when modeling route choices of car drivers, but there could be a behavioral interpretation for the preference for overlapping routes as suggested already by Bovy and Hoogendoorn-Lanser (2005).

A negative PSC parameter suggests that travelers in the multimodal public transport network prefer to travel via routes with high overlap with competing ones, most likely because this corresponds to having more opportunities to reach their destination from their origin. Given the inherent risk of delays and irregularity in public transport networks, travelers might simply value the availability of a large number of en-route alternative options over the uniqueness of the route. Accordingly, in the multimodal public transport network under scrutiny the PSC can be seen as a measure of robustness of the trip by the traveler.

7 Conclusions

This paper has presented the solution to a number of challenges in modeling route choice behavior on a multimodal large-scale public transport network, specifically overcoming the limitations of existing literature either focusing on SP data, RP data concerning only hub-and-spoke networks, or RP data not examining the variety of alternative routes that exist in grid networks. By generating on average 40 alternatives for 5,641 observations, this paper provides insight into the rates of substitution of in-vehicle travel time and other time components while taking a bird's-eye view on the entire door-to-door experience of travelers in a complex network. When considering that the sample is representative of the Danish population thanks to the construction of the TU survey, this is the first route choice model in public transport that depicts the preferences of travelers in a multimodal large-scale public transport network.

Several specifications of the utility function were examined and rates of substitution were calculated for both PSC-Logit and Mixed PSC-Logit models. The most significant outcomes from the model estimates are: (1) the importance of considering heterogeneity across travelers for better representing route choice behavior of public transport users; (2) the relevance of explicitly accounting for the frequency of the services to obtain more accurate rates of substitution between the various time components of the multimodal trip; (3) the significance of examining trip purposes and trip length in the estimation of models for better explaining traveler behavior; (4) the negative PSC parameter implying that public transport travelers prefer overlapping routes.

Although estimates are obtained from a sample of public transport users, and hence a portion of the public with a more positive attitude to public transport than the general population, the representativeness of the sample makes the rates of substitution extremely valuable for policy makers interested in designing multimodal networks responding to the needs of the (potential) users as well as to retain existing users.

Overall, this study tackled satisfactorily the challenges inherent to modeling route choices in a complex multimodal network by successfully applying state-of-the-art choice set generation methods and route choice models. Estimates confirmed the difference in the preferences for in-vehicle travel times of bus, metro, S-trains and other trains, as well as for various components of transfers and access/egress. Moreover, estimates emphasized the importance of considering trip purpose and inserting trip length in the specification of the utility function. At an operator level, punctuality and the estimated in-vehicle value of times seem to be related. However, disaggregate punctuality data on lines, date, and time of day were not available for model estimation purposes.

The results showed the importance of coordination between the different public transport modes in terms of minimized transfer walking and waiting time, the relevance of transfer locations that allows seamless transfer from one vehicle to another, and the significance of easy availability of access and egress modes for example in terms of parking availability for both cars and bicycles. The parameters estimated in the study both provided parameters to enlighten the necessary

improvements in the public transport network to increase the attractiveness of public transport options and provided input to the Danish National Transport Model.

Future research could look more into the concept of estimating weights for public transport route choice by considering hyperpath network (see, e.g., Kurauchi et al. 2012). Additional future work on the route choice model estimation will be to test the parameters through assignments in a large-scale network as the Danish National Transport Model. It is very interesting to obtain more knowledge about the parameters discussed above such as reliability of transport modes or network attributes to include these in the models. Finally, the possibility of collecting data from GPS devices combined with travel diaries would be interesting to touch upon.

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